A First Course in Abstract Algebra

Section 37: Applications of the Sylow Theory

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

Contents

37 Applications of the Sylow Theory 2

Exercise 37.1: Conjugacy classes of the square-symmetry group 2

Exercise 37.2: Composite orders below 60 2

Exercise 37.3: True or false: applications of Sylow theory 3

Exercise 37.4: Groups of order $5 \cdot 7 \cdot 47$ 3

Exercise 37.5: No simple group of order 96 4

Exercise 37.6: No simple group of order 160 4

Exercise 37.7: An order-15 subgroup in every order-30 group 4

Exercise 37.8: Cycle type and partitions 4

Exercise 37.9: Conjugacy classes and class equation of S_4 5

Exercise 37.10: Class equations of S_5 and S_6 5

Exercise 37.11: Conjugacy classes and abelian p-groups 5

Exercise 37.12: The center of S_n for $n > 2$ 6

37 Applications of the Sylow Theory

Exercise 37.1: Conjugacy classes of the square-symmetry group

(a) Write $D_4 = \{e, r, r^2, r^3, s, rs, r^2s, r^3s\}$ with $srs = r^{-1}$. Conjugating a rotation by s reverses it; conjugating a reflection by r advances its exponent by 2. The classes are therefore

$$\{e\}, \quad \{r^2\}, \quad \{r, r^3\}, \quad \{s, r^2s\}, \quad \{rs, r^3s\}.$$

These are disjoint and account for all eight elements. In the textbook's labels the last two classes are the two axis reflections and the two diagonal reflections, while $r = \rho_1$. **(b)** Their sizes give the class equation $8 = 1 + 1 + 2 + 2 + 2$. The two singleton classes are exactly the center $\{e, r^2\}$.

Exercise 37.2: Composite orders below 60

Plan. For completeness, organize the finite list instead of making an unsupported assertion for each order. If a Sylow number is 1, its subgroup is normal. A nontrivial p-group has nontrivial center and, if its order is composite, is not simple. The prime-power composite orders are

$$4, 8, 9, 16, 25, 27, 32, 49.$$

For the following remaining orders, the indicated prime has a unique Sylow subgroup, directly from $n_p \mid |G|/p^a$ and $n_p \equiv 1 \pmod p$:

p	$\lvert G\rvert$
3	$6, 18, 54$
5	$10, 15, 20, 35, 40, 45, 50$
7	$14, 21, 28, 42$
11	$22, 33, 44$
13	$26, 39, 52$
17	$34, 51$
19	$38, 57$
23	46
29	58
11	55

(The rows may use any prime giving uniqueness.) This leaves $12, 24, 30, 36, 48, 56$.

- For 24 and 48, n_2 is 1 or 3. If it is 3 and G were simple, its nontrivial conjugation action on the three Sylow subgroups would embed G into S_3, impossible by orders.
- For 12 and 36, n_3 is 1 or 4. In the latter case simplicity would embed G into S_4. The sign map must then be trivial (a simple group of order greater than 2 has no nontrivial map to $\mathbb{Z}_2$), so G embeds into A_4. Order 36 is too large; at order 12 this would give $G \cong A_4$, which has its normal order-4 subgroup of double transpositions.
- For 30, unless a Sylow 3- or 5-subgroup is normal, there are ten order-3 subgroups and six order-5 subgroups. Their nonidentity elements are disjoint, requiring $10 \cdot 2 + 6 \cdot 4 = 44 > 29$ nonidentity elements.

- For 56, n_7 is 1 or 8. In the latter case the order-7 subgroups already occupy 48 nonidentity elements. Only seven nonidentity elements remain. Every Sylow 2-subgroup has exactly seven nonidentity elements, all among these remaining ones; hence all such subgroups coincide and the Sylow 2-subgroup is normal.

This exhausts every composite order below 60. The conjugation actions used above are nontrivial and, under the assumption of simplicity, have trivial kernel, justifying the embeddings.

Exercise 37.3: True or false: applications of Sylow theory

(a) True. $159 = 3 \cdot 53$, and $53 \not\equiv 1 \pmod 3$; the two-prime criterion gives a cyclic group. Explicitly $n_{53} = 1$ and $n_3 \mid 53$, $n_3 \equiv 1 \pmod 3$ force $n_3 = 1$, so the normal factors commute and their product is cyclic.

(b) True. $102 = 2 \cdot 3 \cdot 17$, and $n_{17} \mid 6$, $n_{17} \equiv 1 \pmod{17}$ force a normal order-17 subgroup.

(c) False. $\mathbb{Z}_6$ is solvable but 6 is not a prime power.

(d) True. The normal chain of Exercise 36.21 has prime-order factors, so a finite p-group is solvable.

(e) This is a judgment about the method rather than a formal theorem; "true" describes the intended observation. Individual Sylow counts and class equations for many distinct orders become lengthy, and some cases require additional arguments.

(f) True. For order 21, $n_7 \mid 3$ and $n_7 \equiv 1 \pmod 7$ give $n_7 = 1$.

(g) True. Its center has order a positive multiple of 5 by the class equation for 5-groups, so it contains at least five elements.

(h) True. For order 42, $n_7 \mid 6$ and $n_7 \equiv 1 \pmod 7$ give $n_7 = 1$.

(i) False. A subgroup order must divide 42, but $8 \nmid 42$.

(j) False. There are other simple groups, for example $\mathrm{GL}(3, \mathbb{F}_2)$ of order $(8-1)(8-2)(8-4) = 168$, which is not of prime order or the order of an alternating group. One way to check simplicity is to count its conjugacy classes: their sizes are $1, 21, 56, 42, 24, 24$ (the nonidentity element orders are respectively $2, 3, 4, 7, 7$). No union of these classes containing the identity has a proper nontrivial size dividing 168. A normal subgroup is a union of conjugacy classes and its order divides 168, so none exists. The class sizes follow from centralizer orders $8, 3, 4, 7, 7$, respectively, by counting invertible matrices commuting with representatives.

Exercise 37.4: Groups of order $5 \cdot 7 \cdot 47$

Apply Sylow to each prime. The possible counts satisfy

$$n_{47} \mid 35, \quad n_{47} \equiv 1 \pmod{47}; \qquad n_7 \mid 235, \quad n_7 \equiv 1 \pmod 7.$$

The divisors of 35 are all below 47, so $n_{47} = 1$. The divisors of $235 = 5 \cdot 47$ are $1, 5, 47, 235$, with residues $1, 5, 5, 4$ modulo 7, so $n_7 = 1$. Also $n_5 \mid 329$; its possible divisors $1, 7, 47, 329$ have residues $1, 2, 2, 4$ modulo 5, forcing $n_5 = 1$. Thus all three Sylow subgroups are normal, cyclic, and of pairwise coprime orders. Two distinct ones commute: a commutator of their elements belongs to both normal

subgroups, whose intersection is trivial by Lagrange's theorem. Their product has all $5 \cdot 7 \cdot 47$ elements and is the direct product of three cyclic groups of pairwise coprime orders. Hence it is cyclic and therefore abelian.

Exercise 37.5: No simple group of order 96

A Sylow 2-subgroup has order 32, and $n_2 \mid 3$, so $n_2 = 1$ or 3. If $n_2 = 1$, it is a proper nontrivial normal subgroup. If $n_2 = 3$, conjugation on the three Sylow subgroups defines a nontrivial homomorphism $G \to S_3$. Were G simple, the kernel, a normal subgroup, would be trivial because the action is nontrivial. This would embed 96 elements into the six-element group S_3, impossible. Thus the kernel is itself a nontrivial proper normal subgroup, proving nonsimplicity in this case too.

Exercise 37.6: No simple group of order 160

Since $160 = 2^5 \cdot 5$, the Sylow 2-number divides 5 and is odd, so it is 1 or 5. The first case gives a normal subgroup of order 32. In the second case the conjugation action gives a nontrivial homomorphism $G \to S_5$. Its kernel cannot be trivial, because an injective map would require $160 \le |S_5| = 120$; and it is not all of G, since the action is nontrivial. Thus it is a nontrivial proper normal subgroup. Both cases exclude simplicity.

Exercise 37.7: An order-15 subgroup in every order-30 group

The counting argument of Example 37.12 shows that an order-30 group has a normal subgroup N of order 3 or of order 5: otherwise ten order-3 subgroups and six order-5 subgroups would require 44 nonidentity elements. If $|N| = 3$, the quotient G/N has order 10, and Cauchy's theorem gives a subgroup Q of order 5. Its preimage under the quotient map has order $3 \cdot 5 = 15$. If $|N| = 5$, the quotient has order 6 and has an order-3 subgroup; its preimage has order $5 \cdot 3 = 15$. Preimages of subgroups are subgroups, so either alternative proves the assertion. The order computation counts $|N|$ elements in each coset of the preimage.

Exercise 37.8: Cycle type and partitions

(a) The permutation $\tau\sigma\tau^{-1}$ sends $\tau(a_i)$ to $\tau(a_{i+1})$, with the last sent to the first, and fixes points outside this cycle. Thus it is $(\tau(a_1) \ldots \tau(a_m))$. **(b)** For two cycles of the same length, define τ to send their entries in cyclic order to one another and extend this bijection arbitrarily to the other points. Part (a) gives the desired conjugation. **(c)** For products of disjoint cycles with the same lengths, do the same on each disjoint support and combine these bijections. Conjugation respects products, so it sends the first product to the second. **(d)** Conversely conjugation only relabels entries and therefore preserves all cycle lengths, including fixed points as length-1 cycles. Thus conjugacy classes correspond exactly to unordered lists of positive lengths summing to n, namely partitions of n. **(e)** The numbers for $n = 1, 2, 3, 4, 5, 6, 7$ are

$$1, 2, 3, 5, 7, 11, 15.$$

For example a systematic way to compute them is to let $P(n, k)$ count partitions with parts at most k and use $P(n, k) = P(n, k-1) + P(n-k, k)$, with $P(0, k) = 1$ and negative arguments zero. The two terms distinguish partitions having no part k from those obtained by removing one part k.

Exercise 37.9: Conjugacy classes and class equation of S_4

By Exercise 37.8 there is one class for each cycle type. A class with m_k cycles of length k has size

$$\frac{n!}{\prod_k k^{m_k} m_k!}.$$

To see the denominator, rotate the entries within each length-k cycle in k ways and permute equal-length cycles in $m_k!$ ways; these do not change the permutation. For S_4 the classes are

type	representative	size
1^4	e	1
$2\,1^2$	(12)	6
2^2	$(12)(34)$	3
$3\,1$	(123)	8
4	(1234)	6

Here each row means all permutations of that type. Hence the class equation is $24 = 1 + 6 + 3 + 8 + 6$.

Exercise 37.10: Class equations of S_5 and S_6

Use the cycle-type formula $n!/\prod_k k^{m_k} m_k!$ from Exercise 37.9. For S_5 the complete table is

type	1^5	$2\,1^3$	$2^2\,1$	$3\,1^2$	$3\,2$	$4\,1$	5
size	1	10	15	20	20	30	24

Thus $120 = 1 + 10 + 15 + 20 + 20 + 30 + 24$. For S_6, list the eleven partitions as follows:

type	size	type	size
1^6	1	$2\,1^4$	15
$2^2\,1^2$	45	2^3	15
$3\,1^3$	40	$3\,2\,1$	120
3^2	40	$4\,1^2$	90
$4\,2$	90	$5\,1$	144
6	120		

Consequently

$$720 = 1 + 15 + 45 + 15 + 40 + 120 + 40 + 90 + 90 + 144 + 120.$$

The displayed sums equal the group orders, checking that all types and counts have been included.

Exercise 37.11: Conjugacy classes and abelian p-groups

The structure theorem writes an abelian group of order p^n uniquely up to isomorphism as

$$\mathbb{Z}_{p^{r_1}} \times \cdots \times \mathbb{Z}_{p^{r_t}}, \quad r_i \geq 1, \quad r_1 \leq \cdots \leq r_t, \quad \sum r_i = n.$$

The unordered exponent list is a partition of n, and different lists give nonisomorphic groups by uniqueness of the prime-power decomposition. Thus there are $p(n)$ isomorphism types of abelian groups of order p^n. Exercise 37.8 gives exactly $p(n)$ conjugacy classes in S_n, since cycle lengths provide the same partitions. The shared count comes from the common partition parameter, not from an asserted group isomorphism.

Exercise 37.12: The center of S_n for $n > 2$

Let $\sigma \neq e$, choose i with $\sigma(i) = j \neq i$, and choose k different from both i, j; this is possible because $n > 2$. Conjugation gives

$$\sigma(i\ k)\sigma^{-1} = (j\ \sigma(k)).$$

This transposition cannot equal $(i\ k)$, because its support contains j, which is neither i nor k. Therefore σ does not commute with $(i\ k)$ and is not central. Every nonidentity permutation fails centrality, while the identity commutes with everything. Hence $Z(S_n) = \{e\}$.